NoSmoothNoFootprint

Processed Using

Lapis Version 0.9

July 22, 2025

RunParameters Folder

The RunParameters folder contains files that have the full details of this run, which can be loaded into Lapis to duplicate part or all of this run.

FullParameters.ini

FullParameters.ini contains all parameters used in this run. Loading it will allow you to duplicate it precisely.

DataParameters.ini

DataParameters.ini contains the parameters that describe the input data of the run. Loading it will allow you to process the same data with new options.

ProcessingParameters.ini

ProcessingParameters.ini contains the parameters that describe the method by which the data was processed. Loading it will allow you to process a new dataset with the same methods.

ComputerParameters.ini

ComputerParameters.ini contains the parameters which are specific to the computer on which the run was performed. Loading it will allow you to perform runs on the same computer without needing to re-set them.

ProcessingAndComputerParameters.ini

ProcessingAndComputerParameters.ini contains all parameters in either ComputerParameters.ini or ProcessingParameters.ini. It's like a profile for how you process lidar, which you can apply to new datasets.

Point Filters

It is very common for lidar data files to include a large number of points which are unsuitable for processing, either because they are errors or because you're doing a special run. This page describes the strategies used to exclude those points from the run.

Outlier Filter

Points below -8 meters or above 100 meters were excluded.

Withheld Filter

Points are sometimes marked 'withheld', indicating that a previous piece of software concluded they were unsuitable for use. Withheld points were excluded from this run.

Class Filter

Lapis does not perform point classification. However, it is common for input data to already be classified by some previous software. If this is the case for this data, then the following standard classes were filtered:

Low Point

Water

High Noise

Scan Angle Filter

Aerial lidar sensors sweep back and forth as they emit pulses. The angle that a pulse was fired at is called the scan angle for that pulse. For some applications, points with a high scan angle are undesirable. In this run, points with a scan angle greater than 30 degrees were excluded.

Output Data Characteristics

All metrics were processed at a cellsize of 10 meters.

Where appropriate, the units of all output data are meters.

All output data of this run is in the following coordinate reference system: WGS 84 / UTM zone 10N + NAVD88 height. The full WKT string is provided below for reference:

```
COMPOUNDCRS["WGS 84 / UTM zone 10N + NAVD88 height",
    PROJCRS["WGS 84 / UTM zone 10N",
        BASEGEOGCRS["WGS 84",
            ENSEMBLE["World Geodetic System 1984 ensemble",
                MEMBER["World Geodetic System 1984 (Transit)"],
                MEMBER["World Geodetic System 1984 (G730)"],
                MEMBER["World Geodetic System 1984 (G873)"],
                MEMBER["World Geodetic System 1984 (G1150)"],
                MEMBER["World Geodetic System 1984 (G1674)"],
                MEMBER["World Geodetic System 1984 (G1762)"],
                MEMBER["World Geodetic System 1984 (G2139)"],
                MEMBER["World Geodetic System 1984 (G2296)"],
                ELLIPSOID["WGS 84",6378137,298.257223563],
                ENSEMBLEACCURACY[2.0]],
            UNIT["degree",0.0174532925199433],
            ID["EPSG",4326]],
        CONVERSION["UTM zone 10N",
            METHOD["Transverse Mercator"],
            PARAMETER["Latitude of natural origin",0],
            PARAMETER["Longitude of natural origin",-123],
            PARAMETER["Scale factor at natural origin",0.9996],
            PARAMETER["False easting",500000],
            PARAMETER["False northing",0]],
```

```
    CS[Cartesian,2],
        AXIS["(E)",east],
        AXIS["(N)",north],
        UNIT["metre",1],
    USAGE[
        SCOPE["Navigation and medium accuracy spatial referencing."],
        AREA["Between 126´°W and 120´°W, northern hemisphere between equator and 84´°N, onshore and offsh
        BBOX[0,-126,84,-120]],
    ID["EPSG",32610]],
VERTCRS["NAVD88 height",
    VDATUM["North American Vertical Datum 1988"],
    CS[vertical,1],
        AXIS["gravity-related height (H)",up],
        UNIT["metre",1],
    USAGE[
        SCOPE["Geodesy, engineering survey, topographic mapping."],
        AREA["Mexico - onshore. United States (USA) - CONUS and Alaska - onshore - Alabama; Alaska; Arizo
        BBOX[14.51,172.42,71.4,-66.91]],
    ID["EPSG",5703]]]
```

Ground Model Algorithm

In this run, Lapis did not calculate its own ground model. Instead, the user supplied a rasterized ground model produced by previous software. The elevation of the ground beneath each point was estimated with a bilinear interpolation from those rasters.

Canopy Surface Model

Overall Description

A canopy surface model (CSM) is a raster product representing Lapis' best guess at the height of the canopy at each point in the area. The CSM files can be found in the CanopySurfaceModel directory. They have names like "NoSmoothNoFootprint_CanopySurfaceModel_Col1_Row1_Meters.tif".

CSM Creation Algorithm

The CSM for this run was produced with the Max Point algorithm. In this algorithm, the value of a cell in the output raster is equal to the maximum height of lidar returns that fall in that cell.

Canopy Metrics

The metrics in the CanopyMetrics directory are derived products from the canopy surface model. Each one is calculated by applying a summary statistic to the canopy pixels contained in the larger metric pixel.

NoSmoothNoFootprint_MaxCSM_Meters.tif

The maximum value attained in the CSM. The units are meters.

NoSmoothNoFootprint_MeanCSM_Meters.tif

The mean of the values of the CSM. The units are meters.

NoSmoothNoFootprint_StdDevCSM_Meters.tif

The standard deviation of the values of the CSM. The units are meters.

NoSmoothNoFootprint_RumpleCSM.tif

The surface area of the CSM, divided by the area of the ground beneath it. A larger value indicates a more complex canopy. This metric is unitless.

Tree-Approximate Objects

Tree-approximate objects (TAOs) represent Lapis’ best guess at where individual trees are. The name TAO reflects the uncertainty inherent in this task.

High Points TAO Identification Algorithm

This run used the High Points algorithm to identify the locations of TAOs. This is a canopy surface model-based algorithm; it it calculated entirely on the CSM, without reference to the original lidar points. A pixel in the CSM is considered a TAO if it is higher than the eight pixels surrounding it. This algorithm works well with tree species, such as conifers, that have a well-defined top.

Watershed Segmentation Algorithm

A tree segmentation algorithm takes the TAOs identified by the identification algorithm and assigns regions of the acquisition to each one. The watershed segmentation algorithm is a canopy surface model-based algorithm; it is performed entirely on the CSM, without reference to the original point data. It gets its name from hydrology, where it is used to segment landscapes into watersheds. The algorithm turns the canopy upside-down and treats it as if it were terrain; each TAO is assigned to the ’watershed’ it would belong to if this imaginary terrain were filled with water.

Products

The output TAO data can be found in the TreeApproximateObjects directory. There are three kinds of products: the TAOs themselves, the segment rasters, and the max height rasters. To avoid unusably large filesizes, they are tiled. Their filenames indicate the row and olumn each tile belongs to. The location of each tile is available in TileLayout.shp, in the Layout directory.

TAOs

The TAOs themselves are stored in files with names like

"NoSmoothNoFootprint_TAOs_Col1_Row1.shp". These are point vector files, whose locations represent the TAOs identified by the identification algorithm. There are six attributes in the attribute table. ID is a unique identifier for each TAO. X and Y are the coordinates of the TAO. Height is the height of the TAO, measured from the canopy surface model, in meters. Area is the area of the region assigned to each TAO by the segmentation algorithm, in square meters. Radius is the estimated crown radius of the TAO, derived directly from the area, as if the TAO was a circle, in meters.

Segments

The files with names like "NoSmoothNoFootprint_Segments_Col1_Row1.tif" represent the division of the landscape into TAOs. There are two variants, raster and polygon. The raster variant assigns each pixel's value to be the ID of the TAO it belongs to, or nodata if it doesn't belong to any TAO. The rasters have the same resolution as the canopy surface model. The polygons represent the same data, converted into polygons,each TAO receiving a unique polygon.

Circles

Max Height

The files with names like "NoSmoothNoFootprint_MaxHeight_Col1_Row1_Meters.tif" are similar to the segments files, but instead of using the TAO's ID as their value, they use the TAO's height. These are thus similar in concept to a canopy surface model, but as if trees were a constant height, instead of having varying heights throughout their area.